

Approach to Kidney Replacement Therapy

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Individuals with advanced chronic kidney disease (CKD) will have one of three fates: stable CKD, progressive loss of kidney function to kidney failure, or death prior to developing kidney failure. The options for treatment for people who progress to kidney failure include kidney replacement therapy (KRT), consisting of transplantation or dialysis, and supportive (nondialysis) care (see [Chapter 94](#)).

Broadly there are two types of dialysis: peritoneal dialysis (PD) and hemodialysis (HD).

The number of people starting dialysis for KRT each year varies enormously around the globe ([Fig. 95.1](#)).¹ The incidence per million population in most countries has risen steadily over the past decade ([Fig. 95.2](#)), although some have seen a decline in recent years ([Fig. 95.3](#)). The incidence of dialysis for KRT is influenced by multiple factors: the incidence and prevalence of diseases that may lead to kidney failure, especially diabetes; the ability of health care systems to identify CKD and to slow progression to kidney failure^{2,3}; the level of kidney function at which KRT is begun; and the availability of resources to provide dialysis.

Dialysis is costly and time-consuming; once started, it may continue for many years. All individuals likely to reach kidney failure, their families, and their caregivers require physical and psychological preparation, including education about future treatment options, given in a form that they find accessible.

PREDICTION OF KIDNEY FAILURE

Preparation for kidney failure treatment requires two things: identification of those individuals who are at high risk for reaching kidney failure and prediction of the likely time when KRT may be needed.⁴

Diabetes, heavy proteinuria, declining estimated glomerular filtration rate (eGFR), and previous episodes of acute kidney injury make it more likely that an individual will progress to kidney failure. A variety of risk prediction tools can be used. The Kidney Failure Risk Equation (KFRE) performs reasonably well over 2 years and has been extensively studied and validated in a variety of populations ([Fig. 95.4](#)).⁵ Models making predictions at 5 years or more considerably overestimate the risk of kidney failure because of the competing risk of death.⁶ The Grams model, which accounts for this, is suitable for predictions over 4 years.⁷

Predicting when someone may need KRT is made easier by a graphical display of eGFR. The trajectory of the eGFR accurately reflects changes in the true GFR over time⁸ and is easier to understand than columns of figures. The graph should be shared with patients to help

them understand how their disease is progressing. If individuals with a declining trend in eGFR are systematically identified,⁹ the number starting KRT without adequate preparation may be reduced and the rising trend in KRT incidence slowed.²

MULTIDISCIPLINARY CARE IN ADVANCED CHRONIC KIDNEY DISEASE

Advanced CKD care aims to address several issues: preservation of remaining kidney function; prevention or treatment of complications of CKD; involvement of the individual, their family, and caregivers in making an informed choice in regard to PD, HD, and conservative kidney management; creation of dialysis access in good time; and, in appropriate individuals, preparation for kidney transplantation ideally before dialysis is started. People who receive consistent predialysis care have better outcomes¹⁰ and incur lower health care costs.

People need time (often months) to understand and make decisions about dialysis and its implications. The best approach is to transfer the care of individuals with a high risk of kidney failure to a multidisciplinary team at least 12 months before dialysis is started. One method of determining who should be referred to multidisciplinary care is to use a risk-based assessment such as the Kidney Failure Risk Equation described earlier. A risk of kidney failure of more than 10% over 2 years has been suggested as a trigger for referral. Alternatively, one can estimate the start date for KRT by extrapolating a trendline of the eGFR forward until it reaches 10 mL/min/1.73 m².¹¹ Using the 12-month prediction rather than an arbitrary value of eGFR to prompt referral avoids older individuals with slowly declining GFR being referred for dialysis preparation unnecessarily.

The trajectory of the eGFR graph may not be linear; intercurrent illness in individuals at stage 5 CKD can cause a sudden drop in GFR and precipitate urgent dialysis. It is therefore prudent to refer all individuals with CKD for multidisciplinary care once the eGFR reaches 15 mL/min/1.73 m².

Predialysis Education Programs

Conventional office consultations with a nephrologist may not enable people with advanced CKD to gain sufficient knowledge and understanding to make good decisions about KRT.¹² Individuals receiving additional care from a predialysis multidisciplinary team have better biochemical results, are more likely to start dialysis in a planned way with less hospitalization, and may even have improved survival rates once they have started dialysis. As well as being good clinical practice,

Global Incidence of Treated Kidney Failure, Based on Individual Country Data

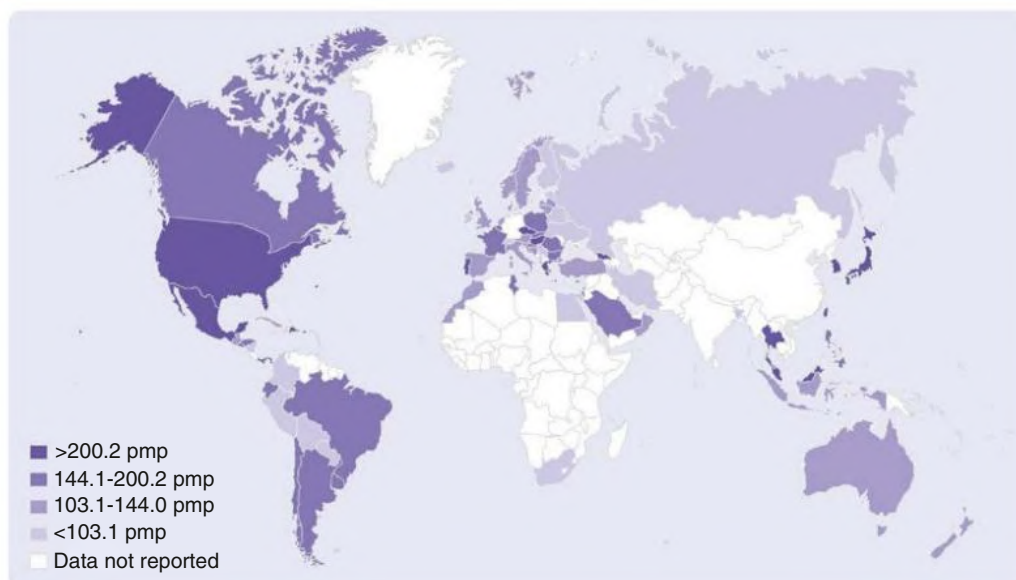


Fig. 95.1 Geographic Variation in the Incidence of Treated Kidney Failure per Million Population (pmp), by Country, 2019. Data presented only for countries from which relevant information was available. All rates are unadjusted. (From Bello AK, Levin A, Lunney M, et al. Status of care for end stage kidney disease in countries and regions worldwide: international cross sectional survey. *BMJ*. 2019;367:l5873.)

Countries or Regions With the Largest Percentage Increase in Incidence of ESKD, 2009–2010 vs 2017–2018

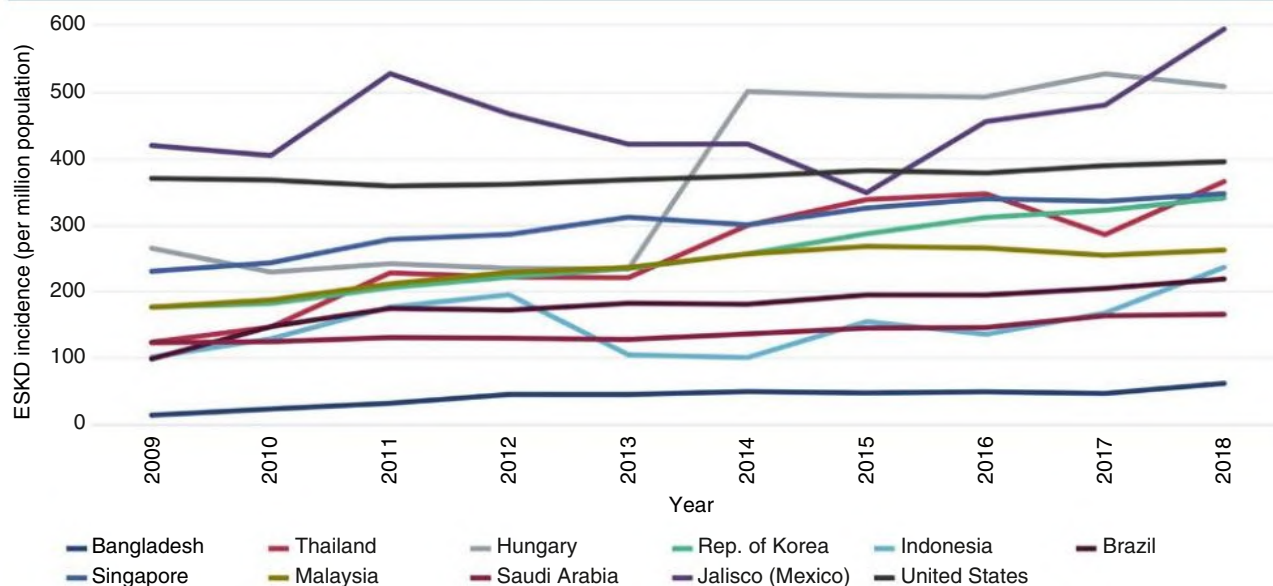


Fig. 95.2 Ten Countries Having the Highest Percentage Increase in the Incidence of End-Stage Kidney Disease (ESKD) From 2009 to 2018. All data are unadjusted. Data for United States are shown for comparison purposes. (From US Renal Data System. 2020 *USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States*. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2020.)

these programs may make good financial sense because the savings in inpatient costs can exceed those required to run the clinics.¹³

Although clinicians are expert in the technical aspects of dialysis and transplantation, people living with CKD know best about their own needs and preferences and should be encouraged to share in making decisions.

Education should follow the principles of adult learning: first, assess the individual's existing level of knowledge and understanding; second, build on this knowledge by delivering appropriate information in an appropriate form; and third, establish that they have understood and accepted the information given. Education can be delivered both

Average Yearly Change in ESKD Incidence Data, 2009–2018

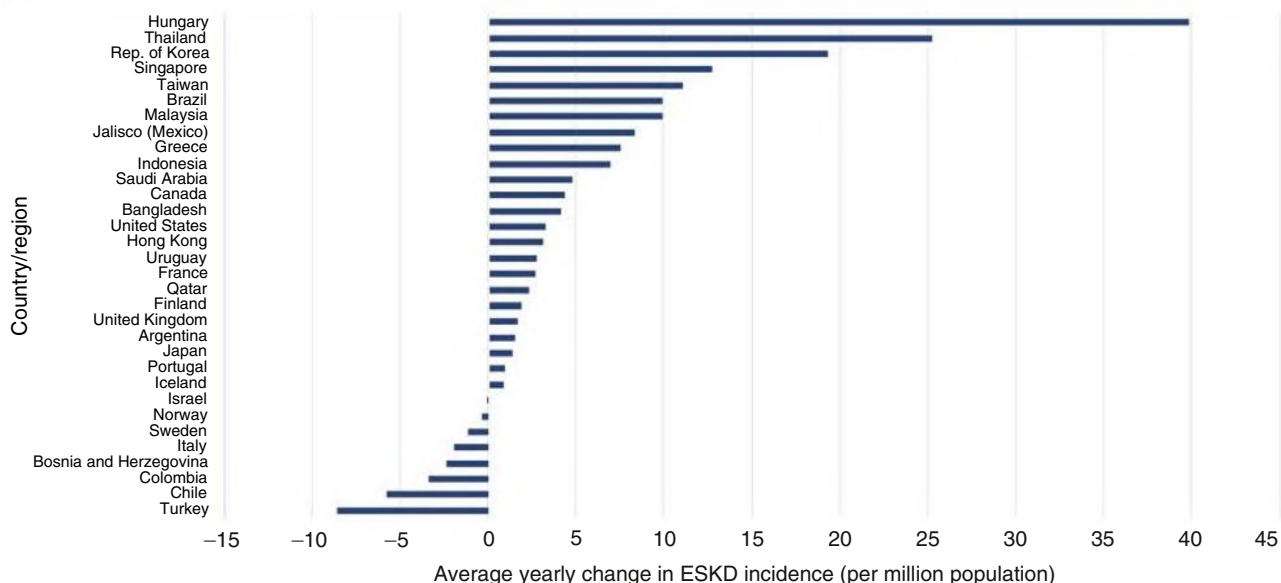


Fig. 95.3 Average Yearly Change in End-Stage Kidney Disease (ESKD) Incidence Data From 2009 to 2018. All data are unadjusted. (From US Renal Data System. 2020 *USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States*. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2020.)

Predicting the Probability of Kidney Failure

Variables	Patient A	Patient B
Age	70 years old	50 years old
Sex	Male	Male
Estimated glomerular filtration rate	30 mL/min/1.73 m ²	30 mL/min/1.73 m ²
Urine albumin to creatinine ratio	200 mg/g	50 mg/g
Albumin	3.5 g/dL	4.0 g/dL
Bicarbonate	21 mEq/L	26 mEq/L
Calcium	9.0 mg/dL	9.8 mg/dL
Phosphorus	4.5 mg/dL	3.8 mg/dL
Kidney failure risk	9.2% (2 years) 27.0% (5 years)	3.6% (2 years) 11.3% (5 years)

Fig. 95.4 Predicted Probability of Kidney Failure for Two Patient Profiles Using the 8-Variable Kidney Failure Risk Equation. The Kidney Failure Risk Equation uses eight common variables to estimate the 2- and 5-year risk of kidney failure.

individually and in groups. In a group education session, individuals living with CKD may learn more from fellow participants than from the group's facilitator. Furthermore, support groups help individuals and their relatives appreciate that they are not alone in facing the demands of kidney failure.

The multidisciplinary team may include a dietician, nurse educator, pharmacist, physical therapist, occupational therapist, social worker, and sometimes a trained peer-support volunteer. There is no ideal team composition, and it will vary depending on the local context.¹⁴ A controlled trial in California studied the value of social worker input to the predialysis program in reducing unemployment.¹⁵ In the intervention group, individuals and their relatives met regularly with a licensed social worker both before and after starting dialysis to explore strategies for continuing their current employment. Blue-collar workers in the intervention group were 2.8 times more likely to continue working. People in employment had a better quality of life, greater self-esteem, and a more positive attitude toward work. Because it is difficult for individuals treated with dialysis to regain jobs once they are lost, this result is particularly valuable for long-term rehabilitation.

Individuals living with CKD should be directed to the wide range of educational materials available. Many national organizations, such as Kidney Care UK (<https://www.kidneycareuk.org/>),¹⁶ provide web-based information and produce printed and audiovisual material. Decision aids can help people think through their options and choose the treatment that suits them best. A good online example is My Kidneys, My Choice (<http://mychoice.kidney.org.au>).¹⁷ Each consultation should be followed by a letter or report, ideally written directly to the patient¹⁸ and copied to their family doctor.

Education About Transplantation

For many, transplantation offers the best prospect for improved survival and quality of life, especially in younger people. Even in older people with greater comorbidity, transplantation can improve survival and be cost-effective.¹⁹

The options of transplanting kidneys from a deceased or living donor should be discussed, as well as combined kidney and pancreas transplants for some people with diabetes. Although outcome data from the local transplant center should be made available, published data can be used, such as from the US Scientific Registry of Transplant Recipients²⁰ or NHS Blood and Transplant in the United Kingdom (<https://www.odt.nhs.uk>).

The ideal time for the transplant to be performed is before dialysis has begun (preemptive transplantation). This avoids the need for access surgery, and in most studies, survival, graft survival, and acute rejection rates are better.²¹

WHEN SHOULD DIALYSIS BE STARTED?

There is no single measure that can be used to determine the right time to start dialysis. A low eGFR, rising serum phosphate, and falling serum bicarbonate may indicate kidney failure, but these levels are also affected by muscle mass and protein intake. Falling serum albumin may indicate inflammation rather than malnutrition secondary to uremia.

The mean eGFR at the start of dialysis varies across countries. In the United Kingdom, the mean eGFR at dialysis initiation was 7.4 mL/min/1.73 m² in 2018. Registry data from USRDS shows 10.6% of individuals starting dialysis with eGFR greater than 15 mL/min/1.73 m² and 27.3% with eGFR 10 to 14 mL/min/1.73 m².²⁰ In Canada, 7.9% started dialysis with eGFR greater than 15 mL/min/1.73 m² and 19.4% with eGFR 10.5 to 15 mL/min/1.73 m².²² The median eGFR at dialysis

initiation in Australia was 7.4 mL/min/1.73 m² and 5.6 mL/min/1.73 m² in New Zealand.²³

In the United States, dialysis was started at a consistently lower eGFR within the Department of Veterans Affairs, a non-fee-for-service health system, than with other providers.²⁴

Limitations of a Purely Clinical Approach to the Initiation of Dialysis

Waiting for uremic symptoms to develop before starting KRT carries the risk that the individual will be malnourished when they start dialysis, with an increased risk for mortality. The chronic nature of progressive kidney disease means that people may remain unaware of the severity of their illness. Protein intake may fall spontaneously so that symptoms of uremia do not develop, at the expense of a loss of lean body mass. Similarly, people may gradually reduce their activities as their exercise tolerance declines. Many individuals only appreciate how ill they have become once they have rehabilitated on a dialysis program. Lack of awareness can be avoided by seeking insidious symptoms of uremia. For example, people should be asked to compare their current eating habits and level of activity with those 6 to 12 months previously. Close friends and relatives may provide a useful third-party view of their well-being. A validated patient-reported outcome measure (PROM) may help identify changing symptoms as CKD advances.²⁵

Limitations of a Purely Laboratory Results–Based Approach to the Initiation of Dialysis

Starting dialysis early must have proven benefits to justify the additional inconvenience, risk of dialysis-related complications, and cost. Dialysis treatment has a finite life, either from loss of peritoneal function or failure of HD access; thus, starting treatment earlier will bring forward the time when further procedures or a change of modality are needed.

Moreover, there is likely to be resistance from many individuals to the suggestion that they should start dialysis if they have no symptoms of uremia. Starting dialysis is the first step in a lifelong commitment to KRT, and people treated with dialysis will be asked to comply with a wide variety of inconvenient and sometimes unpleasant treatments. A high level of compliance is required for a successful outcome. Missed dialysis treatments are associated with increased risk of mortality and health care interventions.^{26,27}

A randomized trial comparing individuals who started dialysis at eGFR of 9.0 mL/min/1.73 m² with those who started 6 months later at eGFR of 7.2 mL/min/1.73 m² showed no difference in survival or quality of life (Fig. 95.5).²⁸ Starting dialysis earlier was not associated with better quality of life and incurred higher health care costs.²⁹ The one advantage in the earlier-start group was that a higher proportion of those who had previously chosen PD actually started dialysis with PD (80% vs. 70%, $P = .01$).

Some clinicians advocate an incremental start to hemodialysis. Incremental hemodialysis is the delivery of less than the conventional thrice-weekly sessions of dialysis with adjustments of dialysis prescription informed by changes in residual kidney function.³⁰ Prescription changes can include frequency of dialysis sessions, session duration, fluid volumes, needle size, dialyzer surface area, blood flow rate, dialysate flow rate. Additionally, dietary intervention can be modified. People with preserved residual kidney function (>500 mL/day), limited weight gain between dialysis sessions (<2.5 kg), and the ability to maintain metabolic parameters in an acceptable range with shorter or fewer sessions would be good candidates for consideration of incremental dialysis. Close monitoring of residual kidney function, interdialytic weight gain, and metabolic parameters are needed to ensure that continuing incremental dialysis is still appropriate. In low-income countries, less frequent dialysis may be all that is available.³¹

CHOICE BETWEEN PERITONEAL DIALYSIS AND HEMODIALYSIS

The mix of PD and HD in the dialysis population varies considerably across countries (Fig. 95.6). In the most recent Global Kidney Health

Atlas, only 32% of low-income countries report that PD is available³² compared with greater than 90% of upper-middle- and high-income countries. Some countries such as Hong Kong and New Zealand have consistently high incidence and prevalence rates of PD.^{33,34} In Hong Kong, PD comprises approximately 75% of prevalent dialysis³³ and since 1985 has been provided as first-choice KRT unless absolute medical contraindications to PD are present.³⁵

Globally, higher expenditure of gross domestic product (GDP) on health care, a larger proportion of private health-care expenditure, and lack of local production of PD consumables and high duty imports on PD supplies are associated with higher costs of PD relative to HD.³⁶ However, in Hong Kong PD has been shown to be more cost-effective than HD.³⁷

Most people with kidney failure are suitable for treatment with either PD or HD. How should we choose between the two? Ideally, we would apply evidence from a randomized clinical trial (RCT) of PD versus HD, but executing such a trial would be difficult. One attempted RCT, the Netherlands Cooperative Study on the Adequacy of Dialysis (NECOSAD), had to be stopped because of low recruitment.

Retrospective and prospective nonrandomized comparative studies have failed to indicate a consistent survival advantage for either modality.³⁸ Previous evidence suggested that PD may be inferior to HD over the longer term in individuals with coronary heart disease and congestive heart failure.³⁹ More recent evidence suggests that this may not be the case,⁴⁰ and PD is associated with superior health-related quality of life.⁴¹

PD technique failure can be due to infection, mechanical problems, inadequate dialysis, and social reasons. Late technique failure is associated mainly with social factors.⁴² Technique failure due to infection and social reasons are associated with a higher short-term risk of death,⁴³ most frequently because of cardiac mortality when technique failure is due to infection, and dialysis withdrawal due to social reasons.

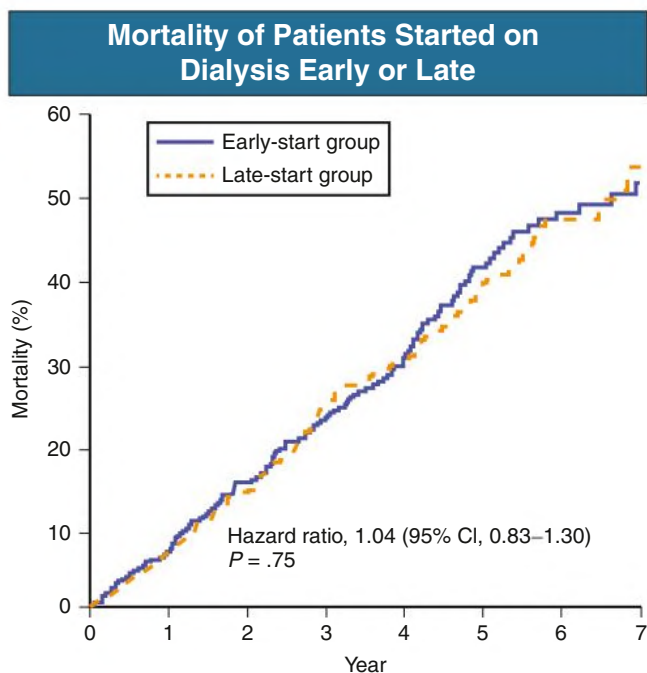


Fig. 95.5 Mortality of Individuals Started on Dialysis Early or Late. Kaplan-Meier curves for the time to death. (From Cooper BA, Branley P, Bulfone L, et al. A randomized, controlled trial of early versus late initiation of dialysis. *N Engl J Med*. 2019;36[7]:609–619.)

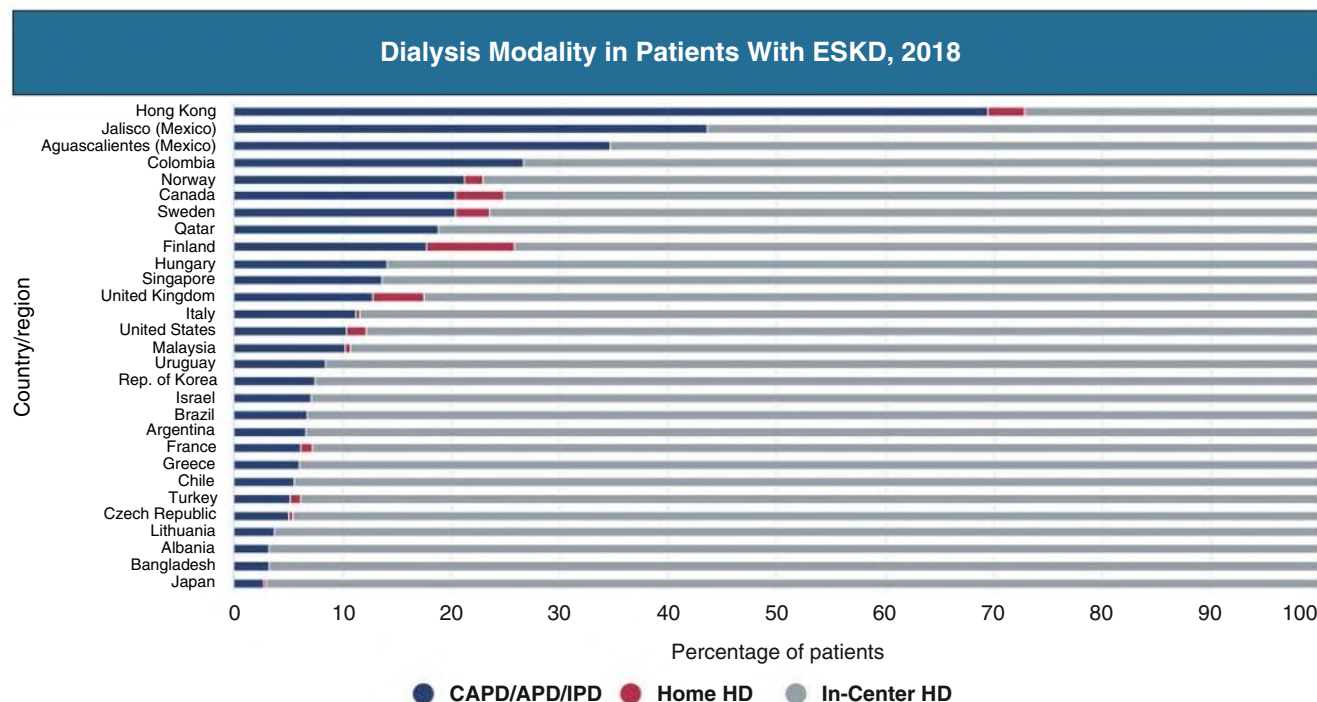


Fig. 95.6 Distribution of the Percentage of Prevalent Patients Using In-Center Hemodialysis (HD), Home HD, or Peritoneal Dialysis (PD), 2018. Denominator is calculated as the sum of patients receiving HD, PD, or home HD; does not include patients with other/unknown modality. APD, Automated peritoneal dialysis; CAPD, Continuous ambulatory peritoneal dialysis; IPD, Intermittent peritoneal dialysis. (From US Renal Data System. 2020 *USRDS Annual Data Report: Epidemiology of Kidney Disease in the United States*. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2020.)

Contraindications to Peritoneal Dialysis

There are a few situations in which PD is contraindicated (Box 95.1). Relative contraindications to PD include those discussed in the following section.

Fresh Intraabdominal Foreign Body

Individuals with prosthetic aortic grafts have been successfully treated with PD. HD is usually used for up to 16 weeks to allow the graft to be covered with epithelium and so avoid the risk for graft infection via peritoneal dialysate. However, this risk must be balanced against that of bacterial seeding from the HD access.

Body Size Limitations and Intolerance of Intraabdominal Fluid Volume

Both very large and very small body size can be challenging for people treated with PD. Small individuals may be intolerant of the volume of dialysate needed to achieve adequate dialysis, particularly if they have negligible residual kidney function. Alternative methods of fluid exchange, such as automated PD, can be used to overcome this limitation. Larger dwell volumes or automated PD are required to achieve adequate solute clearance and ultrafiltration in people with a body mass index greater than 35 kg/m². Discomfort as a result of increased intraabdominal volume can be significant in those with chronic respiratory disease, low-back pain, or large polycystic kidneys. In general, it is hard to predict someone's tolerance of intraabdominal fluid, and so these limitations usually appear after PD has been started.

Bowel Disease and Other Sources of Infection

The presence of ischemic bowel disease, inflammatory bowel disease, or diverticulitis is likely to increase the incidence of peritonitis caused by organisms passing through the bowel wall into the peritoneum. Abdominal wall infection may lead to peritonitis via the exit site and catheter tunnel.

Testing for methicillin-resistant *Staphylococcus aureus* (MRSA) before all elective surgical procedures is good practice. Clearance of nasal *S. aureus* with topical mupirocin cream reduces the risk for staphylococcal infection at the exit site.

Severe Malnutrition or Severe Obesity

Severe malnutrition may lead to poor wound healing and leakage from the catheter tunnel. In addition, peritoneal protein losses during dialysis may exacerbate hypoalbuminemia. At the other end of the spectrum, it may prove difficult to satisfactorily place a peritoneal catheter through the abdominal wall in people with severe obesity, and the risk for infection is greater. Thereafter, absorption of glucose from the dialysate, which may average as much as 800 calories per day, may contribute to further weight gain.

Contraindications to Hemodialysis

Contraindications to HD are few (see Box 95.1). As discussed in Chapter 96, access to the circulation usually can be obtained, even in people with extensive vascular disease or previous surgery. An aversion to needle puncture of the atrioventricular fistula (AVF) is common in the early stages but usually can be overcome by careful use of local anesthetic and nursing encouragement. Severe coagulopathy may make management of anticoagulation in the extracorporeal circuit difficult.

HOME HEMODIALYSIS

In the 1960s, maintenance dialysis was mostly done by the patient and the family at home. Because in-center HD programs were established

BOX 95.1 Contraindications to Dialysis Modalities

Peritoneal Dialysis

Absolute

- Lack of housing/lodging
- Loss of peritoneal function producing inadequate clearance
- Adhesions blocking dialysate flow
- Surgically uncorrectable abdominal hernia
- Abdominal wall stoma
- Diaphragmatic fluid leak
- Inability to perform exchanges in absence of suitable assistant

Relative

- Recent abdominal aortic graft
- Ventriculoperitoneal shunt
- Intolerance of intraabdominal fluid
- Large muscle mass
- Extreme obesity
- Severe malnutrition
- Skin infection
- Bowel disease

Hemodialysis

Absolute

- No vascular access possible

Relative

- Difficult vascular access
- Needle phobia
- Cardiac failure
- Coagulopathy

Modified from Peritoneal Dialysis Adequacy Work Group. Clinical practice guidelines for peritoneal dialysis adequacy. *Am J Kidney Dis.* 2006;48(suppl 1):S91–S97.

and then PD became available, people could choose not to have HD at home. This removed the burden of dialysis from the individual and family and avoided the cost of installing a dialysis machine with its associated water treatment. In the United Kingdom the percentage of the adult dialysis population having home HD fell from 35% in 1984 to 1% in 2006.⁴⁴

Home-based HD offers similar clinical outcomes compared with facility-based HD and provides significant benefits: it removes the inconvenience of traveling to and from the dialysis facility, gives individuals the freedom to dialyze at a time that suits them, reduces the cost of nursing and support staff, and gives the individual more opportunities to take part in home and work activities.⁴⁵

People who dialyze at home often perform more than three treatments per week, and some dialyze overnight. More frequent dialysis significantly reduces the time taken to recover after each treatment, reduces dietary restrictions and antihypertensive medications, and leads to improved quality of life and reduced mortality.⁴⁶

In recent years there has been increasing recognition of the benefits of home HD, and this, along with better technology, has increased nephrologists' enthusiasm for it.⁴⁷ In 2018, the proportion of individuals treated with KRT (including transplant) receiving home HD was 2.0% in the United Kingdom,⁴⁴ 2.1% in the United States,²⁰ and 2.6% in Canada; the highest proportion in Europe was 3.45% in Denmark.⁴⁸ Some countries achieve higher uptake: the proportion in Australia was 4.2% and in New Zealand 8.1%.^{19,22}

Hemodialysis or Hemodiafiltration

Hemodiafiltration (HDF) is increasingly used throughout Europe and elsewhere but is not available in the United States because of regulatory restrictions. HDF adds the removal of fluid containing larger molecules by convection to the clearance of smaller molecules by diffusion. For example, β_2 -microglobulin levels are lower in individuals treated by HDF than by HD. In this way, HDF theoretically mimics glomerular filtration more closely and potentially may benefit people with little or no residual kidney function.

A number of trials comparing outcomes with HD and HDF have been conducted, yielding conflicting results. Evidence suggests that high convection volumes (>30 L/wk/m²) need to be used to deliver better outcomes if HDF is selected as the dialysis modality.⁴⁹

CHOICE OF HEMODIALYSIS OR PERITONEAL DIALYSIS

Ideally, everyone with advanced CKD would be given “modality-neutral” counseling and allowed to select their preferred mode of treatment. When provided with modality education, people are more likely to select and be treated with PD.⁵⁰ Independent predictors for choosing PD include being married, being counseled before the start of dialysis, and increasing distance from the base unit; predictors for choosing HD are increasing age, comorbidity, and male sex. However, although 45% of patients may choose PD, not all of them start dialysis on this modality. Individuals who require urgent dialysis often receive HD rather than PD, although PD can be started urgently if the necessary resources are available.⁵¹ Once started on HD, only a small proportion transfer to PD, even if this was their original preference. Individuals who do transfer from HD to PD are more likely to have technique failure in the following 12 months.⁵² Patients presenting late

for dialysis can start PD urgently,⁵³ and if they do start hemodialysis, they still should have access to an education program about their options for long-term treatment.

Frail individuals can be assisted to have PD at home by a spouse, family member, or trained assistant. This can avoid them having the burden of in-center HD and improve their quality of life,⁵⁴ and they may progress to independent PD after a month or two of assistance.⁵⁵

The major differences in dialysis modality across countries (see Fig. 95.6) suggest that the type of dialysis individuals receive is more often determined by physicians and organizational factors than by their preferences.

IMPORTANCE OF DIALYSIS ACCESS

Before starting dialysis, every patient would ideally have made an informed choice between PD and HD after weeks or months of counseling and preparation, and those choosing HD would have a functioning AVF. Unfortunately, dialysis is frequently started in less than ideal circumstances. The percentage of individuals presenting shortly before starting dialysis varies between dialysis units and countries (Fig. 95.7). This is a particular issue in the United States, where in 2015 only 31% of individuals beginning KRT had received 12 months of nephrology care before KRT.⁵⁶

People presenting to a nephrologist shortly before starting dialysis have a longer initial hospital stay and a higher incidence of major complications and death.¹⁰ Late presentation is one reason why HD is started using a catheter rather than an AVF (Fig. 95.8), and increased catheter use is associated with increased mortality.

The timing of AVF placement is complex.⁵⁷ If an AVF is created many months before the individual is predicted to reach kidney failure, then the chances that a catheter is used are reduced. However,

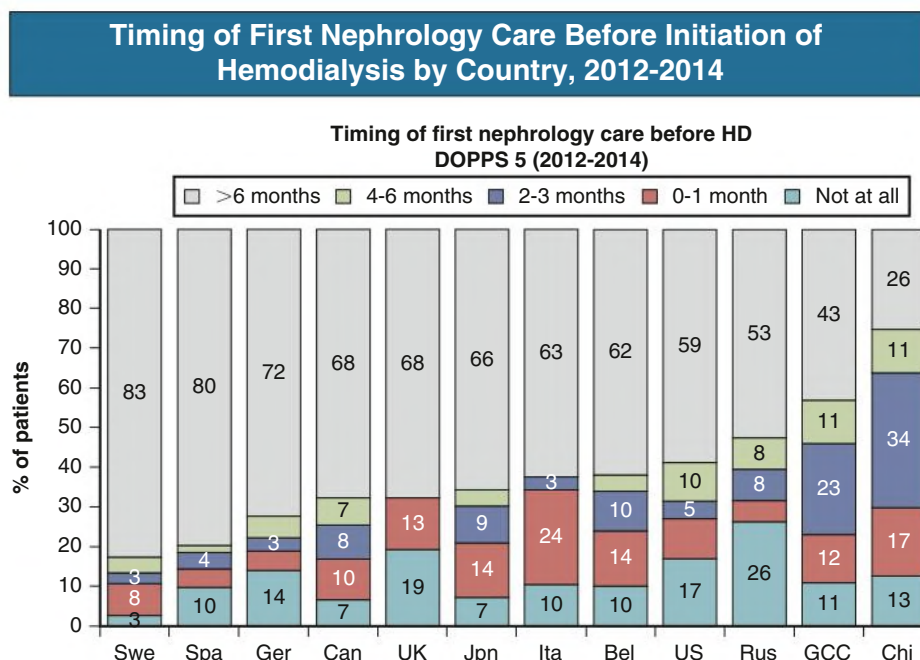


Fig. 95.7 Global Differences in Timing of First Nephrologist Care Before Starting Dialysis Among Individuals Enrolled in the Dialysis Outcomes and Practice Patterns Study (DOPPS) Within 60 Days of Starting Dialysis (2012–2014). Data were calculated among individuals on dialysis for fewer than 60 days at the time of enrollment in the study. *Bel*, Belgium; *Can*, Canada; *Chi*, China; *Ger*, Germany; *GCC*, Gulf Cooperation Council countries; *HD*, hemodialysis; *Ita*, Italy; *Jpn*, Japan; *Rus*, Russia; *Spa*, Spain; *Swe*, Sweden. (From Pisoni RL, Zepel L, Port FK, Robinson BM. Trends in US vascular access use, patient preferences, and related practices: an update from the US DOPPS Practice Monitor with international comparisons. *Am J Kidney Dis*. 2015;65[6]:905–915.)

Type of Vascular Access in Use Among Incident Hemodialysis Patients by Country, 2012-2014

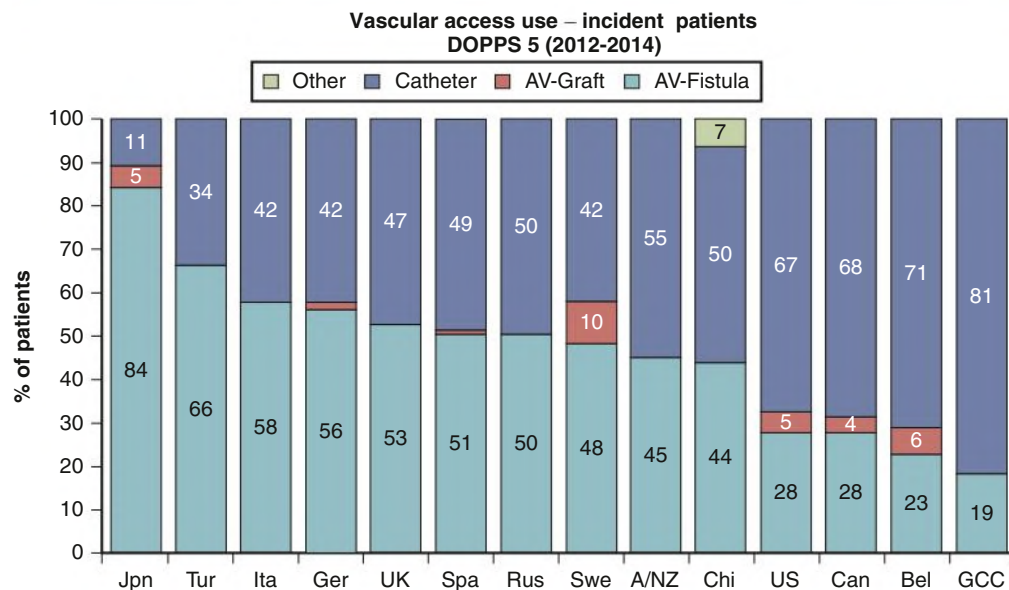


Fig. 95.8 Country Differences in Timing of First Nephrologist Care Before Starting Dialysis Among Individuals Enrolled in the Dialysis Outcomes and Practice Patterns Study (DOPPS) Within 60 Days of Starting Dialysis (2012–2014). Data were calculated among individuals on dialysis for fewer than 60 days at the time of enrollment in the study. AV, Arteriovenous; *Bel*, Belgium; *Can*, Canada; *Chi*, China; *Ger*, Germany; *GCC*, Gulf Cooperation Council countries; *Ita*, Italy; *Jpn*, Japan; *Rus*, Russia; *Spa*, Spain; *Swe*, Sweden. (From Pisoni RL, Zepel L, Port FK, Robinson BM. Trends in US vascular access use, patient preferences, and related practices: an update from the US DOPPS Practice Monitor with international comparisons. *Am J Kidney Dis*. 2015;65[6]:905–915.)

elderly individuals may die without the AVF ever being used.⁵⁸ AVF placement after dialysis has started is associated with a lower chance of long-term fistula use.⁵⁹ International data from the Dialysis Outcomes and Practice Patterns Study show that the risk for death is higher for individuals dialyzing in units that use central venous catheters in a high proportion of patients.⁶⁰ The risk is also increased if arteriovenous grafts are commonly used. Catheters can cause complications such as infection, pulmonary embolism, and central venous stenosis.

There is wide variation across countries in the time required for permanent access to be created and used. If a fistula can be created swiftly and used as soon as it is mature, perhaps after only 2 weeks, the need for a catheter may be avoided. Details of HD access surgery are discussed in [Chapter 96](#) and PD catheter placement in [Chapter 101](#).

DECISION WHETHER TO OFFER KIDNEY REPLACEMENT THERAPY

Availability of Dialysis Facilities

Because of its high cost, KRT is not available to most people with kidney failure worldwide. The practice of rationing dialysis has been candidly documented in a report from a South African center,⁶¹ in which more than half the people with kidney failure assessed between 1988 and 2003 were not offered dialysis. Socioeconomic factors such as age, race, employment, and marital status outweighed medical factors in the decision to begin treatment.

The phenomenon of “supply-sensitive care” applies to dialysis as it does to health care in general.⁶² KRT incidence is independently associated with a country’s GDP, the percentage of GDP spent on health care, and the reimbursement rate relative to GDP for HD

dialysis facilities. In more developed countries, those with a greater proportion of private for-profit facilities have a higher incidence of dialysis.⁶³

Selection by Physicians and Nephrologists

Although incidence rates of KRT in most countries have increased in recent decades (see [Fig. 95.2](#)), not all of this is due to increases in the incidence of kidney failure.^{63,64} In other words, physicians have become more willing over time to offer dialysis to people with kidney failure. The practice of starting dialysis in people who are very elderly and dependent on others for their care or who have multiple comorbid conditions varies significantly across countries and among nephrologists within those countries.⁶⁵ In the UK only 19% of individuals on hemodialysis were fully independent. In Japan in 2010, 13% of individuals treated with hemodialysis had dementia, and 54% of these individuals spent more than half the day in bed.⁶⁶

Uncertainty about prognosis, lack of palliative care services and training, and financial incentives toward providing dialysis may persuade nephrologists to offer dialysis.⁶⁷

Rationing Versus Rational Dialysis Treatment

How dialysis should be used has raised a range of bioethical issues over the years.⁶⁸ In the 1960s, committees would meet to decide who should be offered dialysis, on the grounds that the greatest good should be derived from the limited resources available. Individuals who were expected to survive for only a few months would not be offered dialysis. As resources became more widespread, a rationing approach became unacceptable and was replaced by decisions based on the balance of benefit and harm gained from dialysis for each individual.

Predictive Factors

Are there objective criteria that can be applied to identify individuals for whom the harms of dialysis will outweigh the benefits? One criterion to be dismissed is age. Although advanced age was used as a simple exclusion criterion in the early days of dialysis, the elderly are now the most rapidly growing section of the dialysis population.

There are several mortality risk prediction scores that estimate 6-month mortality for individuals starting dialysis.⁶⁹ For older individuals, a high Ramipril Efficacy in Nephrology (REIN) score predicts 6-month mortality with high sensitivity and specificity.⁷⁰

ADVISING ABOUT PROGNOSIS ON DIALYSIS

Despite these uncertainties, individuals should be given an estimate of their likely future on dialysis. The Renal Physicians Association (RPA) has suggested the following criteria to help identify those over 75 years of age who have a poor prognosis on dialysis⁷¹: (1) clinicians' response of "No, I would not be surprised if my patient died in the next 6 months"; (2) high comorbidity score; (3) significantly impaired functional status (e.g., Karnofsky Performance Status score <40); and (4) severe chronic malnutrition (i.e., serum albumin <2.5 g/dL using the bromocresol green method). Quality of life is also strongly predictive of mortality, even after statistical correction for these comorbid factors.

For individuals whose prognosis is particularly uncertain, or where there is disagreement between the views of the individual and the dialysis team, a time-limited trial of dialysis may be offered. This may give the individual and their family a better understanding of what life on dialysis entails and allow time for further discussion between all parties. The duration of the trial should be judged for each person, and clinical and biochemical parameters such as serum albumin reviewed regularly.

The RPA in the United States has issued guidelines for decisions to not initiate or to discontinue dialysis (Fig. 95.9) and provides a comprehensive toolkit to support shared decision-making.⁷¹

CONSERVATIVE KIDNEY CARE

Conservative kidney care can be defined as planned, holistic, patient-centered care for individuals with stage 5 CKD including a full range of treatment and support but not dialysis.⁷² This modality is increasingly adopted for a subset of individuals with kidney failure and multiple comorbidities.⁷³ To date, evidence to inform discussions about conservative care as an alternative to dialysis has come from observational studies of varied quality.⁷⁴ A meta-analysis found that at 1-year survival was similar for individuals managed with dialysis (73%) and conservative care (71%), but that at 2 years survival diverged to 62% for the dialysis group and 44% for the conservative care group. The Prepare for Kidney Care trial has been undertaken to address this evidence gap.⁷⁵ It is an RCT comparing preparing for either dialysis or conservative care in frail, multimorbid older individuals with advanced CKD.

When asked in a survey about their priority between quality of life or extending the length of life if they were faced with a serious illness, Europeans strongly favored quality of life, with only a few expressing a preference for extending life.⁷⁶ Individuals who are likely to have an unacceptable quality of life should not be subjected to the discomfort of dialysis. Individuals choosing conservative kidney management are spared surgical procedures and the inconvenience and discomfort of traveling to the HD unit or performing PD. Depending on age and the number of comorbidities, survival with dialysis may be no longer, and much of any additional time is made up of days having or recovering from dialysis treatments.⁷⁷ Quality of life is likely to be better without dialysis, and patients are more likely to die at home.

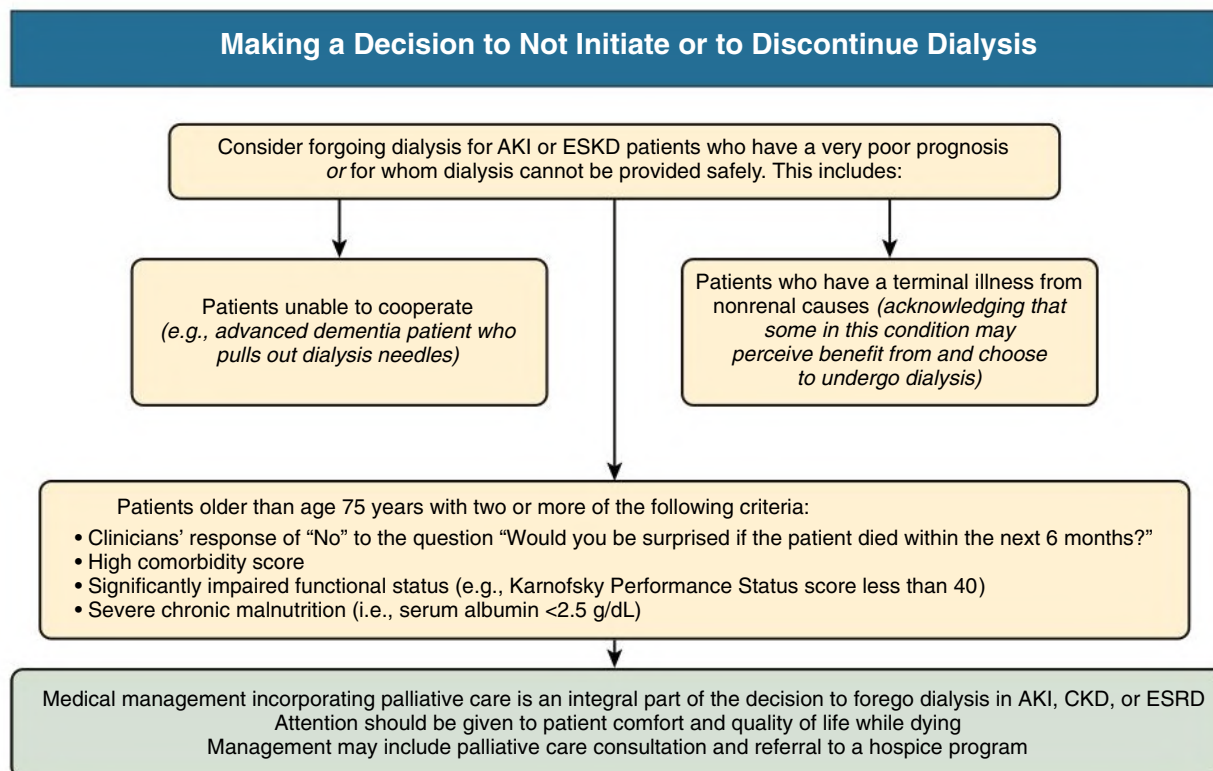


Fig. 95.9 Making a decision to not initiate or to discontinue dialysis. *AKI*, Acute kidney injury; *CKD*, chronic kidney disease; *ESKD*, end-stage kidney disease.

Many of the symptoms and complications of kidney failure, such as anemia, acidosis, itching, insomnia, depression, fluid overload, and hypertension, can be treated with medication and diet. Individuals may not report symptoms such as pain or itching that they do not associate with kidney failure and so should be asked specifically about them.⁷⁸ Symptom management tools and other resources for individuals who chose conservative care and their care providers are available at <https://www.ckmcare.com/>.

Family physicians can be supported to deliver conservative care along with a specialist CKD multidisciplinary team and palliative care specialists.

INDIVIDUALS WHO DO NOT WANT DIALYSIS

Nephrologists may be presented with the dilemma of someone who has decision-making capacity and whom they would normally treat but who does not wish to have dialysis. From an ethical viewpoint, an individual's decision not to start dialysis or to discontinue it is justified on the principle of individual autonomy. Legally, in the United Kingdom this is based on the individual's common law right to self-determination and in the United States on the constitutional right of liberty. Where the individual is able to express a clear wish, the physician is obliged to respect this, because to treat them against their will constitutes an assault.

The physician must nonetheless ensure that all reversible factors have been addressed, such as unfounded fears about what dialysis will entail or a depressive illness affecting the patient's judgment, and ideally request a psychiatric evaluation. It is not uncommon for individuals to express a strong desire not to have dialysis, particularly if they are relatively asymptomatic, only to change their mind when they become more symptomatic. At this late stage, the basic "will to survive" comes to the fore. An advance directive not to have dialysis should never be held as a reason against a change of mind.

DISAGREEMENT ABOUT A DECISION TO DIALYZE

There will inevitably be differences of opinion about the harms and benefits of dialysis to individuals. Dialysis nurses may disagree with the nephrologist's decision to treat someone. If the dialysis nurses and doctors are functioning well as a team, they should feel able to express these reservations and have the issue adequately discussed. It is demoralizing for staff to feel pressured into giving treatment they believe to be inappropriate.

The nephrologist may remain reluctant to offer dialysis despite the insistence of either the individual or, more often, their family or caregivers, the legal agent, or another doctor (see Fig. 95.9). Dialysis must never be given if it is against the individual's clearly expressed wishes, despite the insistence of others. However, if they insist on treatment against the nephrologist's advice, dialysis usually should be given while a resolution is reached.

Extensive discussions and explanations of the treatment options and prognosis may be needed to gain a better understanding of the reasons behind the differing views. Helpful advice may be obtained from another physician, particularly the individual's family doctor, who will have a broader understanding of their circumstances. It may be appropriate to involve a psychologist, social worker, or religious counselor.

It may be necessary to refer the case to a formal ethics committee, if one exists locally, to clarify the issues of disagreement and enable a resolution. A physician cannot be compelled to offer treatment against his or her professional judgment, but the physician is ethically and legally obliged to attempt to transfer the care of the individual to

another physician. Only as a last resort, if no alternative dialysis unit can be found and after adequate notice has been given, should dialysis be withdrawn. The RPA Clinical Practice Guideline Toolkit provides a systematic approach to conflict resolution if there is disagreement regarding the benefits of dialysis⁷¹ (Fig. 95.10).

MANAGEMENT OF DISRUPTIVE INDIVIDUALS ON DIALYSIS

Most nephrologists have had experience of treating a small number of individuals who, for one reason or another, will not comply with the discipline required for maintenance dialysis and who become disruptive to the staff and others. This behavior can range from non-adherence with treatment, which harms the individual but is merely inconvenient to the staff, to verbal or even physical aggression toward the staff and others in the unit. The impact of this small number of individuals can be very great.

The strategy for dealing with this situation must be tailored to the individual, particularly their decision-making capacity. Useful suggestions for resolving conflict have been provided by the RPA (Box 95.2). They emphasize the importance of understanding, information, patience, and persistence. However, the bottom line for individuals who have capacity and are aggressive toward staff on the dialysis unit must be that they are taken off dialysis and sent home.

RESUSCITATION AND WITHDRAWAL OF DIALYSIS

Cardiopulmonary Resuscitation

If people are to be fully involved in making decisions about their treatment, two sensitive issues need to be discussed: cardiopulmonary resuscitation (CPR) and the possibility of withdrawal of dialysis. The two are not necessarily linked; individuals may wish to continue with dialysis but express a desire that resuscitation not be attempted should they have a cardiac arrest. The outcome of CPR in patients treated with dialysis is worse than in the general population, the odds ratio for mortality being 1.24 (95% confidence interval, 1.11–1.3; $P < .001$). In 2011 survival after CPR in the United States was 31%, but a greater proportion of people with kidney failure who survived a cardiac arrest were discharged to skilled nursing facilities.⁷⁹ In Taiwan in 2012, 18.8% of individuals suffering an in-hospital cardiac arrest were discharged from hospital, with a median survival of 6.8 months.⁸⁰ For a witnessed cardiac arrest in an outpatient dialysis unit, fewer than half survive and are admitted to hospital, and only 26% survive to discharge from hospital.⁸¹

A decision not to attempt CPR must be documented in the individual's medical and nursing records, and all nursing staff must be made aware of it. It is important to be clear what is meant by "cardiac arrest" and for there to be agreement on how the nursing staff should respond to a hypotensive "crash" while on dialysis. These notes may form part of an advance directive, as discussed later.

Withdrawal of Dialysis

It is not possible to predict accurately which individuals will gain prolonged benefit from dialysis; thus, many nephrologists offer dialysis to all individuals with kidney failure who wish to have it. This policy ensures that no one is denied dialysis but has the inevitable consequence that some people are started on dialysis who subsequently have an unacceptable quality of life. The possibility of withdrawing dialysis needs to be addressed if these individuals are not to suffer unreasonably.

Rates of withdrawal vary widely between countries and cultures. Withdrawal rates are much higher in Australia, New Zealand, and

Suggested Steps for Resolving Conflict in the Shared Decision About Starting Dialysis

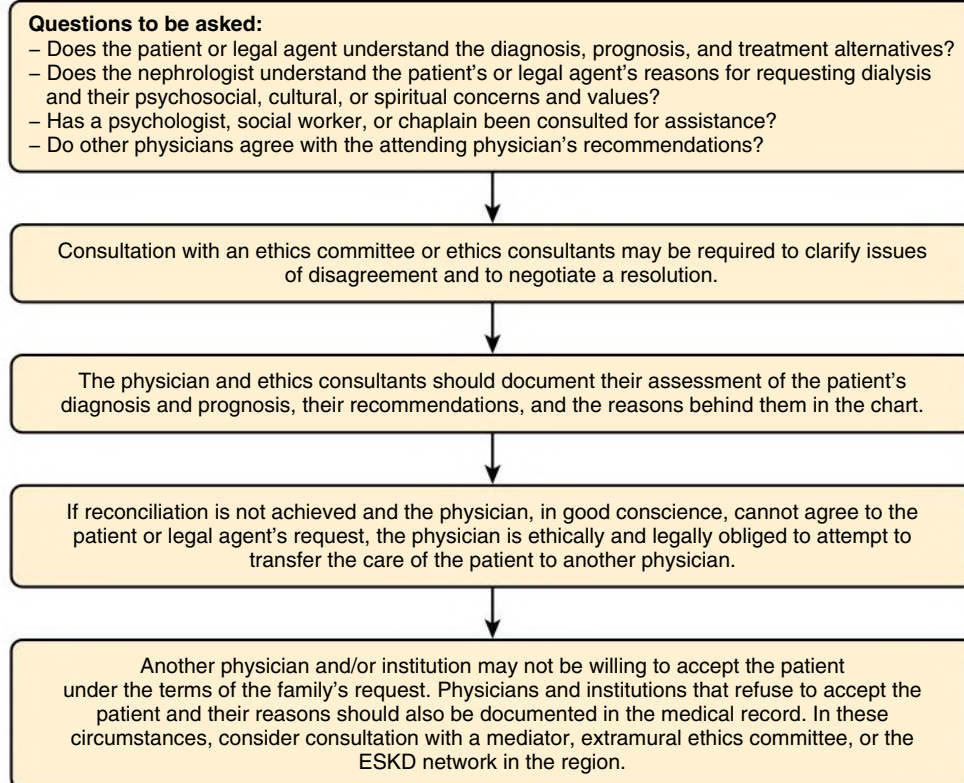


Fig. 95.10 Suggested steps for resolving conflict in the shared decision about starting dialysis. ESKD, End-stage kidney disease. (Modified from Shen JI, Mitani AA, Saxena AB, Goldstein BA, Winkelmayer WC. Determinants of peritoneal dialysis technique failure in incident US patients. *Perit Dial Int*. 2013;33(2):155–166.)

BOX 95.2 Suggested Steps for Dealing With Disruptive Individuals

Identify and document problem behaviors and discuss them with the individual.

- Seek to understand their perspective.
- Identify their goals for treatment.
- Share control and responsibility for treatment with them.
 - Educate them so they can make informed decisions.
 - Involve them in the treatment as much as possible.
 - Negotiate a behavioral contract with them.
- Consult a psychiatrist, psychologist, or social worker for assistance in clinical management or determination of decision-making capacity.
- Be patient and persistent; try not to be adversarial.
- Allow the individual to vent concerns but do not tolerate verbal abuse or threats to staff or others.
- Contact law enforcement officials if physical abuse is threatened or occurs.
- If satisfactory resolution has not occurred with these strategies, contact the local ESKD Network to discuss the situation and ensure due process.
- As a last resort, consider transferring the individual to another facility or discharging them.
- Obtain legal advice before proceeding with plans for discharge, and do not discharge without advance notice and a full explanation of future treatment options.

Modified from Butler CR, Mehrotra R, Tonelli MR, Lam DY. The evolving ethics of dialysis in the United States: a principlist bioethics approach. *Clin J Am Soc Nephrol*. 2016;11(4):704–709.

Canada than in Japan and Italy.⁸² For the countries where withdrawal is more common, the rates are much higher in the 4 months after the start of dialysis. Individuals who withdraw from dialysis are more likely to be older and female, require assistance for transferring and eating, reside in an institution, and have multimorbidity. Rates of dialysis withdrawal are higher in units where conservative care is always or usually discussed, which suggests that individuals' wishes are not always fully included in these decisions.

People may be reticent to express a wish to withdraw from dialysis. Many see it as their duty as a patient to go along with the treatment recommended by their physician and do not wish to appear ungrateful for the efforts that are being made to keep them alive. Their physician may be the last member of the team to learn about their views, and it is very important that good communication exists within the multidisciplinary team so that any clues that the individual gives are acted upon. Staff should adopt a proactive approach and raise the issue of withdrawal of dialysis with individuals who are not thriving.

People's willingness to discuss planning for the end of their life will be affected by factors such as age, ethnicity, cultural background, and religious beliefs, as well as the circumstances of their daily lives. Qualitative research suggests that, when done sensitively, raising the subject of death does not destroy hope for the future. Early discussion of these issues can lead to a more satisfactory outcome for patients, relatives, and staff when the individual eventually dies. In the United States, formal advance directives play an important part in these discussions, and helpful guidance on how to conduct sensitive interviews has been published.⁸³

In the United Kingdom, dialysis teams may enter individuals on an “at risk” register so they can receive appropriate assessment and care. The dialysis unit should have close links with palliative care specialists so there is a smooth transition from maintenance to palliative dialysis, where priority is given to symptoms and person-centered outcomes. Palliative care for people with kidney disease is discussed further in [Chapter 94](#). Individuals who take a long time to recover after a dialysis treatment may benefit from reducing the number of treatments per week.⁸⁴

When someone is no longer competent to make a decision, an advance directive can provide a clear legal basis for the decision to stop dialysis. Indeed, some US states (e.g., Missouri and New York) insist that dialysis must be continued in the absence of such clear and convincing written evidence. In other US states and the United Kingdom, the physician is given the task of deciding on the individual’s behalf. Helpful advice for dialysis staff and individuals wishing to complete an advance directive is available in the RPA Clinical Practice Guideline Toolkit (see [Fig. 95.9](#)).⁷¹

Once someone has expressed a wish for dialysis to be withdrawn, or his or her relatives have raised the issue, the first priority must be to identify any reversible factors that may improve their health sufficiently for the decision to be reversed. In particular, any depression must be identified and treated. Once all these factors have been ruled out, the process of withdrawing dialysis should be managed according to some key principles ([Box 95.3](#)).

BOX 95.3 Key Principles Underlying the Process of Withdrawal of Dialysis

- The ultimate responsibility for the decision rests with the physician, not the relatives or caregivers.
- The individual’s interests and dignity should be protected at all times.
- The process should not be rushed. If there is any doubt about the correctness of the decision, treatment should continue.
- There should be an open discussion among the multidisciplinary team to avoid any damaging disagreements.
- The psychological needs of the health care team should not be overlooked.
- Palliative care must be given in the most appropriate environment (e.g., a hospice or, ideally, the individual’s own home).

Withdrawing dialysis should not be seen as an admission of failure but as a final stage in the process of KRT. The opportunity provided by dialysis to complete unfinished emotional and financial business can make the subsequent bereavement period much less traumatic. Managing this terminal phase can be uniquely rewarding, particularly if it allows an individual and their family and caregivers to prepare themselves for the end of their life.

SELF-ASSESSMENT QUESTIONS

- Which of the following provides the *best* outcomes in someone with advanced CKD?
 - Deceased donor transplant
 - Preemptive living donor transplant
 - Home hemodialysis
 - Peritoneal dialysis
- Early initiation of dialysis is associated with:
 - improved survival
 - improved quality of life
 - decreased morbidity
 - higher health care costs
- Dialysis should be started in someone with advanced CKD when:
 - eGFR is less than 15 mL/min/1.73 m² (CKD stage 5)
 - Urine output is decreased
 - Clinical and biochemical features of azotemia develop
 - Serum bicarbonate is less than 20 mEq/L
- A decision to transfer someone to multidisciplinary renal care should be made when:
 - symptoms of kidney failure begin
 - the eGFR reaches 10 mL/min/1.73 m²
 - the individual is in CKD stage 4 with stable eGFR
 - the trajectory of eGFR indicates a likely need to start kidney replacement therapy within the next 12 months
- The nephrologist may withhold dialysis from someone with end-stage kidney disease:
 - if they have verbally expressed a wish not to have dialysis.
 - if it is against the judgment of the nephrologist.
 - if the individual is disruptive on dialysis.
 - if the dialysis nurses do not wish to continue dialysis.
- Which of the following is *not* predictive of a poor prognosis on dialysis?
 - Karnofsky performance status score less than 40
 - Severe chronic malnutrition
 - Age older than 75 years
 - Physician would not be surprised if the individual died within the next 6 months

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